Overview, roadmap and glossary

QUESTION

What was done in this project, in which order, where each step is explained, and what every recurring term and symbol means.

All reports · Report 01 →

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The project asks how the APL neuron, the single inhibitory neuron of each mushroom body, is wired with the projection neurons (PNs) and Kenyon cells (KCs) in the calyx, and whether that wiring can explain how blocking APL changes the KC odor code in Yang's recordings (TNT, APL blocked, vs the TNT3c control). The work falls into three parts: building the synapse data (Reports 01–03), a model of KCs with APL feedback compared with the recordings (Reports 04–09), and APL's geometry with local, non-spiking versions of the model (Reports 10–12). Each report states its question, inputs, outputs and scripts at the top, defines every symbol in a table, and explains each figure in its caption. All reports exist as HTML (links work, best on screen) and PDF.

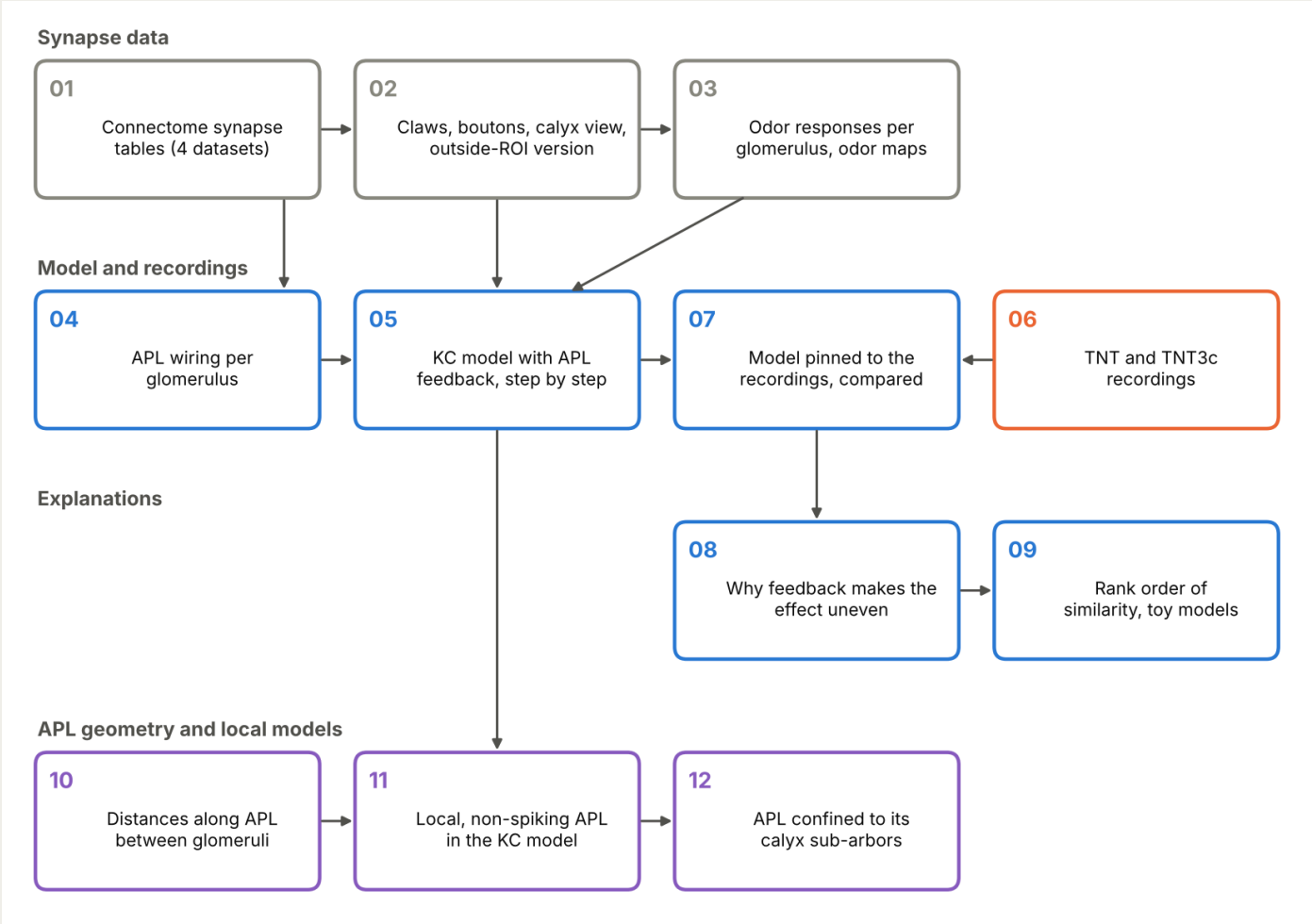


Figure 1. Roadmap of the reports. Each box is one report (number top left). Arrows point from a report to a report that builds on it; reading in number order also works. Not drawn, to keep the figure readable: Report 10 uses the claws and boutons of Report 02, and Report 11 uses the distances of Report 10, the pinning of Report 07 and the rank-order readout of Report 09; Report 12 uses the sub-arbors of Report 10. Box outlines: gray = synapse data; blue = the model and its comparison with the recordings; orange = the recordings; purple = APL geometry and the local models.

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1 The reports in order

Report 01: Connectome data: APL synapse tables and per-cell counts. One table per connectome (FlyWire, hemibrain, male CNS, BANC) with every APL synapse in the mushroom body: who the partner is, where the synapse is, and its region and compartment. Synapses are counted as T-bar → PSD pairs. KCs are APL's main partner everywhere; in the calyx PNs come second. Per KC, synapses onto APL and from APL go together (Spearman 0.62–0.75, BANC 0.46).

Report 02: Claws, boutons, calyx geometry and the outside-ROI version. Boutons (on PN axons) and claws (on KC dendrites) are found by clustering synapse sites within 2 µm; claws are 94–95% from one PN. An APL synapse is placed on a claw or bouton when within 2 µm: almost all APL ↔ PN synapses sit on boutons, but only 18–27% of KC → APL and 30–49% of APL → KC calyx synapses sit on claws. The posterior view used in every map is defined here. The second version of the tables adds synapses just outside the official ROIs, mostly the back of the male CNS calyx (+31%).

Report 03: Odor-weighted maps of APL synapses. Yang's ORN and GH146 responses are averaged over flies per glomerulus and attached to each APL synapse through its PN or claw. An odor's strongest boutons are scattered over the whole calyx. Weighting odor similarity by APL's on-claw synapses barely changes it.

Report 04: APL wiring per glomerulus: the connectome numbers the model uses. Per glomerulus: APL → PN synapses per PN → KC synapse (bouton weight p_g), APL → KC synapses on its claws per claw PN synapse (claw weight q_g), and the boutons' share f of APL's calyx output (about 10%). These are the wiring numbers the model uses.

Report 05: The APL block model, built step by step. The KC model: KC input from claws × glomerulus responses, a threshold, and APL feedback solved self-consistently, in a uniform version and versions using per-KC counts, per-glomerulus bouton and claw weights. Built and explained step by step.

Report 06: The TNT and TNT3c recordings: what is measured. The recordings: 8 TNT and 11 TNT3c recordings (each treated as an independent hemisphere), 16 odors. With APL blocked more units respond and every odor pair becomes more similar (mean change in cosine similarity 0.18).

Report 07: The model pinned to the recordings, and the comparison. The model is pinned to the recordings (each odor's blocked fraction matches TNT; one APL gain matches TNT3c's mean). Uniform feedback predicts which pairs gain similarity (r 0.46 with ORN input, 0.32 with GH146; data ceiling about 0.84). Per-KC counts and boutons at their real share do not beat shuffled wiring. The model's change is about half the data's.

Report 08: Why APL feedback makes the effect of blocking uneven across odors. Why: feedback holds down odors that drive many KCs more than odors that drive few, so blocking APL helps strong odors most, and pairs of strong odors gain the most similarity.

Report 09: Rank order of odor-pair similarity, with toy simulations. Whether inhibition changes the **order** of odor-pair similarities. Global inhibition keeps the order exactly only in special cases; the model keeps it almost (Spearman 0.97–0.98), the data less (0.87, about 0.92 after correcting for noise). Toy simulations show each case.

Report 10: Distances along APL between glomeruli. Distances along APL's skeleton between the microglomeruli of every pair of glomeruli, hemibrain and male CNS. APL's calyx arbor comes in sub-arbors that join only farther down, so glomeruli in the same sub-arbor are much closer along APL.

Report 11: A local, non-spiking APL in the KC model. APL as a non-spiking neuron: inhibition spreads along its skeleton and fades as $e^{-d/\lambda}$. Short-range inhibition on KCs predicts which odor pairs change most better than a uniform APL (the real wiring beats within-class shuffles in 3 of 4 tests), but barely changes the rank order; inhibition on PN boutons alone comes closest to the data's reordering, and is too weak at its real share.

Report 12: APL confined to its calyx sub-arbors. APL with its signal confined to each calyx sub-arbor of Report 10 (no crossing between them). Rank order stays at about 0.98, like a uniform APL; the pattern of which pairs change is no better than one APL unless each sub-arbor's potential is divided by its cable length, and never better than the local model without walls.

2 Glossary of terms

Table 1. Terms. Symbols are defined anew in each report's own table; the most common ones are below.

Term	Meaning
APL	Anterior paired lateral neuron: one large GABAergic (inhibitory) neuron per mushroom body that receives input from KCs and inhibits them and the PN boutons. Thought to be non-spiking (graded potentials), like the locust GGN.

Term	Meaning
Kenyon cell (KC)	The intrinsic neurons of the mushroom body (about 2,000 per side). Classes αβ, α′β′ and γ by the lobes their axons enter. KCs get odor input in the calyx and send output in the lobes.
Projection neuron (PN)	An antennal-lobe output neuron that carries odor information from one glomerulus (uniglomerular) or several to the calyx and the lateral horn.
ORN	Olfactory receptor neuron. ORNs expressing the same receptor converge on one glomerulus.
Glomerulus	A unit of the antennal lobe (e.g. DM1, VA1v); names follow FlyWire. Each uniglomerular PN belongs to one.
Calyx, peduncle, lobes	Parts of the mushroom body: dendritic input region, axon bundle, and axon output region (α, α′, β, β′, γ lobes).
Compartment	Subdivision of a lobe (α1 … γ5) defined by its dopaminergic and output neurons.
Bouton	A swelling of a PN axon in the calyx holding many release sites; here a cluster of the PN's T-bars within 2 µm.
Claw	A KC dendritic ending that wraps a bouton; here a cluster of the KC's PN-input sites within 2 µm. Its glomerulus is that of its main PN.
Microglomerulus	One bouton with the claws around it, and APL's synapses there.
T-bar, PSD	The presynaptic release site and a postsynaptic density facing it. One T-bar can face several PSDs.
Synaptic weight	Number of T-bar → PSD pairs from one neuron onto another in a region (neuPrint "weight", FlyWire "syn_count").
ROI	Region of interest: a dataset's official 3D volume for a brain region. "Official ROIs" = the main tables; the second version adds synapses just outside them.
Hemibrain, male CNS, FlyWire, BANC	The four electron-microscopy connectomes (Report 01). The model uses hemibrain (right side) and male CNS (both sides).
Posterior view	Calyx seen from behind: fly's right to the right, dorsal up (Report 02, Table 2).
ORN table, GH146	Yang's glomerular responses: ORN imaging (fly means; 16 odors used by the model) and GH146 PN imaging (peak ΔF/F; 13 of those odors).
TNT, TNT3c	Yang's KC recordings with APL blocked by tetanus toxin (TNT) and the control with an inactive toxin (TNT3c, APL intact).
Recording, unit	One imaged hemisphere (treated as independent), and one imaged KC region of interest in it.
Responding, fraction responding	A unit responds to an odor when its response exceeds 1 (data); a model KC when its response is above 0. The fraction responding is also called the coding level.
Cosine similarity	Similarity of two odors' population responses: dot product divided by the product of lengths. 1 − cosine is the distance in Yang's RDMs.
ΔS	Change in similarity of an odor pair from intact to blocked (TNT − TNT3c in the data).
Rank order	The ordering of odor pairs by similarity. Measured by the Spearman correlation between blocked and intact similarities over pairs.
Spearman, Pearson	Rank correlation, and linear correlation.
Threshold-linear	A response that is 0 below a threshold and grows linearly above it: $[x-\theta]_+$.
Feedback, self-consistent	APL's activity depends on the KC responses it inhibits, so responses and APL activity are solved together as a steady state.

Term	Meaning
Pinning	Setting each odor's input gain so that the model's blocked fraction responding equals the TNT fraction (Report 07).
Calibration	Choosing one gain (APL strength) by bisection so that a target is met, here the mean intact fraction responding of TNT3c.
Shuffle, null	Rewiring that keeps totals but breaks the real assignment (e.g. KCs swap synapse counts within their class), to test whether the real wiring matters.
Bootstrap	Resampling recordings with replacement to get the spread of a statistic.
Noise ceiling	How well any prediction could match the data given the data's own variability (split-half reliability).
Geodesic distance	Distance along a neuron's skeleton between two points, as opposed to a straight line.
Length constant λ	In a passive neurite, the distance over which a steady signal falls to 37% ($1/e$).
Sub-arbor	A piece of APL's calyx arbor that connects to the others only outside the calyx (Report 10); a compartment in Report 12.

Table 2. Recurring symbols.

Symbol	What it stands for	Values, units, range
$k,\ j$	Kenyon cells	
g	Glomerulus	
$o,\ a,\ b$	Odors (a, b as a pair)	
W_{kg}	Share of KC k's claw PN input from glomerulus g	0 to 1
r_{go}	Response of glomerulus g to odor o (negatives set to 0 in the model)	ORN or GH146 units
g_o	Input gain of odor o (pinning)	
$x_{ko},\ y_{ko}$	Input and response of KC k to odor o	
θ	KC threshold	1 in Reports 07–12
A_o	APL activity for odor o (one-unit APL)	
a	APL gain onto KCs	calibrated
$u_k,\ v_k$	APL → KC and KC → APL calyx synapses of KC k, each ÷ its mean	mean 1
$p_g,\ q_g$	Per-glomerulus bouton and claw weights	Report 04
f	Boutons' share of APL's calyx output synapses	about 0.1
M_{kj}	Coupling from KC j to KC k through APL	Reports 11–12
$d(s,t),\ \lambda$	Distance along APL between sites s and t; length constant	µm
ϕ_o	Fraction of KCs (or units) responding to odor o	0 to 1
$S_{ab},\ \Delta S_{ab}$	Cosine similarity of odors a and b; its change when APL is blocked	
ρ_{rank}	Spearman correlation over pairs between blocked and intact similarity	
r	Pearson correlation over pairs between model and data ΔS	

3 Decisions that apply to every report

- Synapse counts are T-bar → PSD pairs and stand in for synaptic strength.
- Two versions of every table and plot are kept: official ROIs, and with synapses outside them added (Yang, 2026-10-14).
- The model and the comparison with the recordings use only hemibrain and male CNS (Yang, 2026-10-21).
- TNT3c is the control (APL intact) and TNT has APL blocked; each recording is an independent hemisphere (Yang, 2026-10-15).
- Odor-pair similarity is cosine similarity, as in Yang's RDMs (from 2026-10-21).
- Per-KC KC → APL and APL → KC counts in the model are calyx only (from 2026-10-21).